2021 TERM 2 PLANNER FOR YEAR 11 METHODS - SUBJECT TO CHANGE -

METHODS UNITS 1 & 2

Week	Lesson 1 - (MON P1 / MON P1)	Lesson 2 - (MON P2 / MON P2)	Lesson 3 - (WED P5 / WED P6)	Lesson 4 - (FRI P6 / FRI P5)
1 19 Apr	# Polynomials * 7C Ramainder Theorem √	# Polynomials * 7C Factor Theorem √	Trade-Off √	# Polynomials * 7C Rational-root factor theorem Test 2 Review
2 26 Apr	ANZAC Day	ANZAC Day	# Polynomials * 7D Solving cubic equations * 7F Graphs of factorised cubic functions Part 1	# Polynomials * 7E Cubic functions of the form f(x)=a(x-h)^3+k * 7F Graphs of factorised cubic functions Part 2
3 03 May	# Polynomials * 7G Finding rules for cubic functions	# Polynomials * 7I Applications of polynomial functions	# Probability * 9A Sample spaces & probability * 9B Estimating probabilities	# Probability * 9C Multi-stage experiements * 9D Combining events
4 10 May	# Probability * 9E Probability tables * 9F Conditional probability * 9G Independent events	# Probability * Ex 9E - 9G	# Counting Methods * 10A Addition and multiplication principles	# Counting Methods * 10B Arrangements
5 17 May	# Counting Methods * 10C Selections * 10D Applications to probability * 10E Pascal's triangle & the binomial theorem	# Counting Methods * Ex 10C - 10E	Methods Unit 1 Revision (Polynomials)	Methods Unit 1 Revision (Linear & Quadratics) Methods Unit 1 Revision (Transformations of functions)
6 24 May	S1 Exams	S1 Exams	S1 Exams	S1 Exams
7 31 May	S1 Exams	S1 Exams	S1 Exams	S1 Exams
8 7 Jun	WA Day	WA Day	Exam Review (CF)	Exam Review (CA)
9 14 Jun	# Differential Calculus * 17A Average rate of change	# Differential Calculus * 17B Instantaneous rate of change	# Differential Calculus * Ex 17A & 17B	# Differential Calculus * Sadler 5B **CAP: INV 2 MON**
10 21 Jun	# Differential Calculus * 17C The derivative	# Differential Calculus * 17D Rules for differentiation	# Differential Calculus * Sadler 5C &5D	Trade-Off ?
11 28 Jun	# Differential Calculus * 17E Differentiating x^n (n is Z-)	# Differential Calculus * 17F Graphs of the derivative function	# Differential Calculus * 17G Antidifferentiation of polynomial	# Differential Calculus * Sadler Misc 5

YEAR 11 METHODS: POLYNOMIALS

Term 2 Week 2

7C Factorisation of polynomials Part 2:

Rational-root theorem

Learning Objectives/Syllabus Covered:

- 1.1.19 Factorise cubic polynomials in cases where a linear factor is easily obtained

7C Factorisation of polynomials

- Remainder theorem
- Factor theorem
- Rational-root theorem

$$a^2 + 2ab + b^2 = (a+b)^2$$

$$a^2 - 2ab + b^2 = (a-b)^2$$

$$a^2 - b^2 = (a+b)(a-b)$$

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2)$$

$$a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Warm-Up: 5 minutes CF

- Factorise $P(x) = x^3 + 4x^2 + x - 6$

Warm-Up: 5 minutes CF

- Factorise $P(x) = x^3 + 4x^2 + x - 6$

$P(1) = 1^3 + 4 \times 1^2 + 1 - 6 = 0$ ✓

Factor Theorem gives $(x-1)$ is a linear factor of $P(x)$. ✓

$$
\begin{array}{r|l}
 & x^2 + 5x + 6 \\
\hline
x - 1 & x^3 + 4x^2 + x - 6 \\
 & -)\ x^3 - x^2 \\
\hline
 & 5x^2 + x \\
 & -)\ 5x^2 - 5x \\
\hline
 & 6x - 6 \\
 & -)\ 6x - 6 \\
\hline
 & 0
\end{array}
$$

✓

$\therefore P(x) = (x-1)(x^2+5x+6)$

$= (x-1)(x+3)(x+2)$ ✓ ✓

$$
\begin{array}{c|c}
x^2 & 6 \\
\hline
x & 3 \\
x & 2
\end{array}
$$

$3x + 2x = 5x$

Warm-Up: 5 minutes CF

Factors of -6 : $\pm 1, \pm 2, \pm 3, \pm 6$

- Factorise $P(x) = x^3 + 4x^2 + x - 6$

$P(1) = 1^3 + 4 \times 1^2 + 1 - 6 = 0$ ✓

$P(-2) = (-2)^3 + 4 \times (-2)^2 + (-2) - 6$

$= -8 + 16 - 2 - 6$

$= 0$ ✓

$P(-3) = (-3)^3 + 4 \times (-3)^2 + (-3) - 6$

$= -27 + 36 - 3 - 6$

$= 0$ ✓

According to the Factor Theorem:

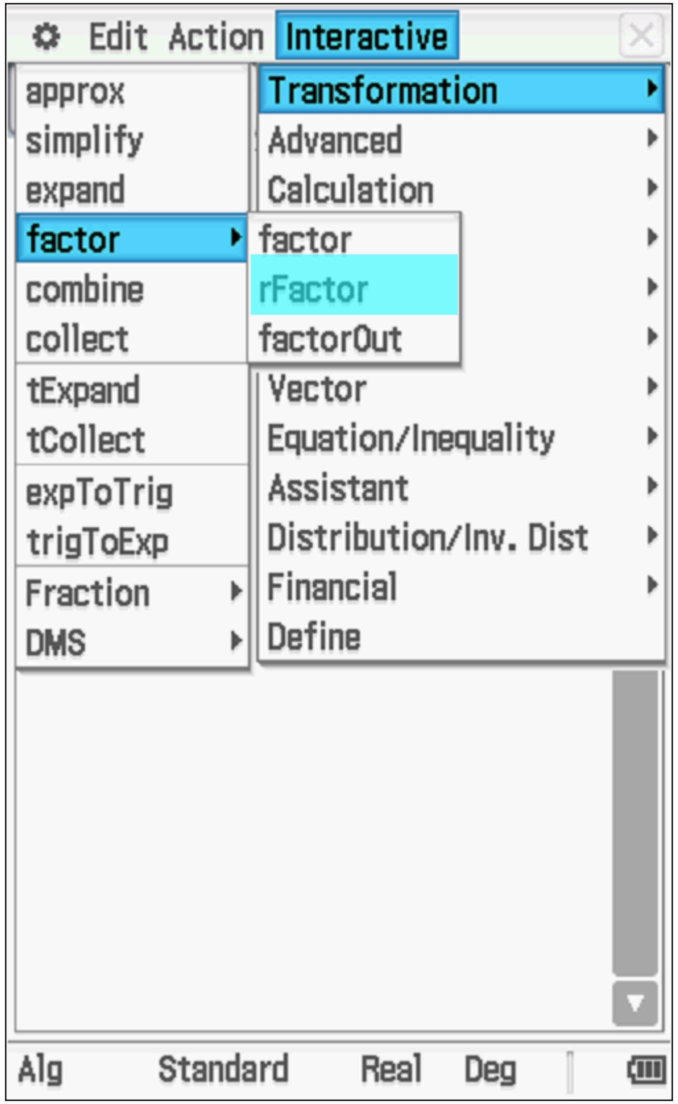
$P(x) = (x-1)(x+2)(x+3)$ ✓ ✓

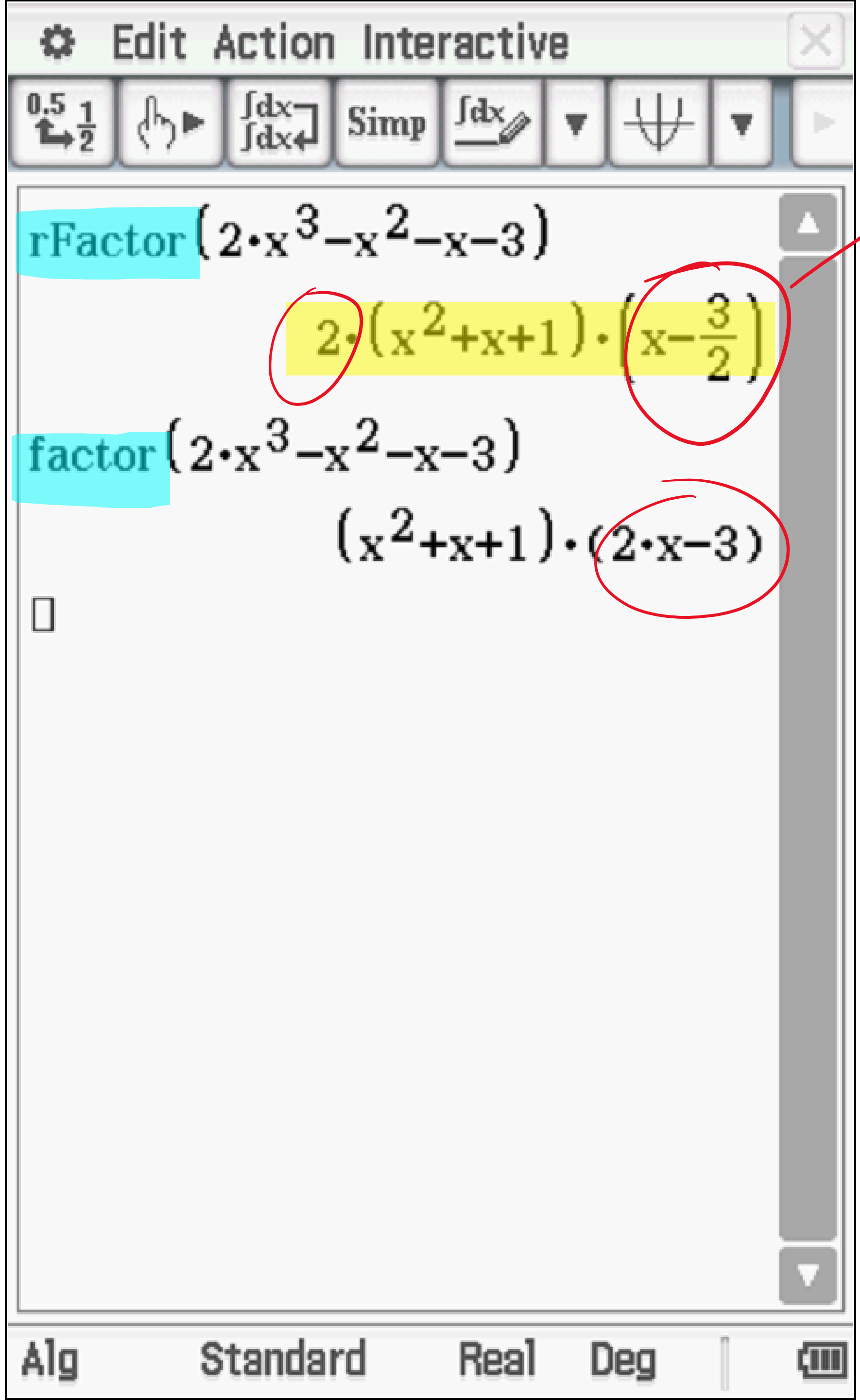
▶ Rational-root theorem

CA : Factorise $2x^3 - x^2 - x - 3$

Rational-root theorem

CA: Factorise $2x^3 - x^2 - x - 3$





$x = \frac{3}{2}$

Showing the actual value of the root

Example 13

Factorise $3x^3 + 8x^2 + 2x - 5$.

Rational-root theorem

Let $P(x) = \overset{\beta}{a_n} x^n + a_{n-1}x^{n-1} + \cdots + a_1 x \overset{\alpha}{+ a_0}$ be a polynomial of degree n with all the coefficients a_i integers. Let α and β be integers such that the highest common factor of α and β is 1 (i.e. α and β are relatively prime).

Example 13

Factorise $3x^3 + 8x^2 + 2x - 5$.

Let $P(x) = 3x^3 + 8x^2 + 2x - 5$

Check that $P(x) = 0$ has no integer solution

Rational-root theorem

Let $P(x) = \overset{\beta}{a_n} x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + \overset{\alpha}{a_0}$ be a polynomial of degree n with all the coefficients a_i integers. Let α and β be integers such that the highest common factor of α and β is 1 (i.e. α and β are relatively prime).

Example 13

Factorise $3x^3 + 8x^2 + 2x - 5$.

Let $P(x) = 3x^3 + 8x^2 + 2x - 5$

Check that $P(x) = 0$ has no integer solution

Here $a_n = 3$, $a_0 = -5$

$\therefore \beta = \pm 3$, $\alpha = \pm 5$ & ± 1

$\therefore x = \frac{5}{3}$ or $-\frac{5}{3}$ or $\frac{1}{3}$ or $-\frac{1}{3}$

Rational-root theorem

Let $P(x) = \overset{\beta}{a_n} x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + \overset{\alpha}{a_0}$ be a polynomial of degree n with all the coefficients a_i integers. Let α and β be integers such that the highest common factor of α and β is 1 (i.e. α and β are relatively prime).

Example 13

Factorise $3x^3 + 8x^2 + 2x - 5$.

Let $P(x) = 3x^3 + 8x^2 + 2x - 5$

Check that $P(x) = 0$ has no integer solution

Here $a_n = 3$, $a_0 = -5$

$\therefore \beta = \pm 3$, $\alpha = \pm 5$ & ± 1

$\therefore x = \frac{5}{3}$ or $-\frac{5}{3}$ or $\frac{1}{3}$ or $-\frac{1}{3}$

Check: $P(\frac{5}{3}) = 3\times(\frac{5}{3})^3 + 8\times(\frac{5}{3})^2 + 2\times(\frac{5}{3}) - 5 = \frac{310}{9}$

$P(-\frac{5}{3}) = 3\times(-\frac{5}{3})^3 + 8\times(-\frac{5}{3})^2 + 2\times(-\frac{5}{3}) - 5 = 0$ $\rightarrow$ $(3x+5)$ is a factor of $P(x)$

$P(\frac{1}{3}) = 3\times(\frac{1}{3})^3 + 8\times(\frac{1}{3})^2 + 2\times(\frac{1}{3}) - 5 = -\frac{10}{3}$

$P(-\frac{1}{3}) = 3\times(-\frac{1}{3})^3 + 8\times(-\frac{1}{3})^2 + 2\times(-\frac{1}{3}) - 5 = -\frac{44}{9}$

Example 13

Factorise $3x^3 + 8x^2 + 2x - 5$.

$(3x+5)$ is a factor of $P(x)$

$$
\begin{array}{r}
x^2 + x - 1 \\
3x+5 \,\big)\, \overline{3x^3 + 8x^2 + 2x - 5} \\
-)\; 3x^3 + 5x^2 \\
\hline
3x^2 + 2x \\
-)\; 3x^2 + 5x \\
\hline
-3x - 5 \\
-)\; -3x - 5 \\
\hline
0
\end{array}
$$

$\therefore P(x) = (3x+5)(x^2+x-1)$ ← Is this fully factorised?

Example 13

Factorise $3x^3 + 8x^2 + 2x - 5$.

$\therefore \; P(x) = (3x+5)(x^2+x-1)$ → Complete the square to factorise this quadratic

$$\left[x^2 + 2 \times \tfrac{1}{2}x + \left(\tfrac{1}{2}\right)^2\right] - \left(\tfrac{1}{2}\right)^2 - 1$$

$$= \left(x + \tfrac{1}{2}\right)^2 - \frac{5}{4}$$ → Use Difference of Two Square Rule

$$= \left(x + \tfrac{1}{2}\right)^2 - \left(\frac{\sqrt{5}}{2}\right)^2$$

$$= \left(x + \frac{1}{2} + \frac{\sqrt{5}}{2}\right)\left(x + \frac{1}{2} - \frac{\sqrt{5}}{2}\right)$$

Hence, $3x^3 + 8x^2 + 2x - 5 = 3\left(x + \frac{5}{3}\right)\left(x + \frac{1+\sqrt{5}}{2}\right)\left(x + \frac{1-\sqrt{5}}{2}\right)$

FULLY FACTORISED !!!

Practice (CF):

Use the rational-root theorem to help factorise each of the following cubic polynomials:

$$P(x) = 2x^3 - 3x^2 - 12x - 5$$

Practice:

Use the rational-root theorem to help factorise each of the following cubic polynomials:

$$P(x) = 2x^3 - 3x^2 - 12x - 5$$

$\beta = \pm 2$, $\alpha = \pm 5$ or ± 1

$\therefore x = \pm \frac{5}{2}$ or $\pm \frac{1}{2}$

Practice:

Use the rational-root theorem to help factorise each of the following cubic polynomials:

$$P(x) = 2x^3 - 3x^2 - 12x - 5$$

$\beta = \pm 2$, $\alpha = \pm 5$ or ± 1

$\therefore\ x = \pm \frac{5}{2}$ or $\pm \frac{1}{2}$

Check: $P(\frac{5}{2}) = -\frac{45}{2}$

$P(-\frac{5}{2}) = -25$

$P(\frac{1}{2}) = -\frac{23}{2}$

$P(-\frac{1}{2}) = 0$

$\therefore$ $(2x+1)$ is a factor of $P(x)$

Practice:

Use the rational-root theorem to help factorise each of the following cubic polynomials:

$$P(x) = 2x^3 - 3x^2 - 12x - 5$$

$\beta = \pm 2$, $\alpha = \pm 5$ or ± 1

$\therefore\ x = \pm \frac{5}{2}$ or $\pm \frac{1}{2}$

Check: $P(\frac{5}{2}) = -\frac{45}{2}$

$P(-\frac{5}{2}) = -25$

$P(\frac{1}{2}) = -\frac{23}{2}$

$P(-\frac{1}{2}) = 0$

$\therefore$ $(2x+1)$ is a factor of $P(x)$

$$
\begin{array}{r|l}
 & x^2 - 2x - 5 \\
\hline
2x+1 & 2x^3 - 3x^2 - 12x - 5 \\
 & -)\ 2x^3 + x^2 \\
\hline
 & -4x^2 - 12x \\
 & -)\ -4x^2 - 2x \\
\hline
 & -10x - 5 \\
 & -)\ -10x - 5 \\
\hline
 & 0
\end{array}
$$

$\therefore\ P(x) = (2x+1)(x^2 - 2x - 5)$

Not fully factorised

Practice (CF):

Use the rational-root theorem to help factorise each of the following cubic polynomials:

$$P(x) = 2x^3 - 3x^2 - 12x - 5$$

$$\therefore P(x) = (2x+1)(x^2 - 2x - 5)$$

$$\longrightarrow (x^2 - 2x + 1) - 1 - 5$$

$$= (x-1)^2 - 6$$

$$= (x-1)^2 - (\sqrt{6})^2$$

$$= (x - 1 + \sqrt{6})(x - 1 - \sqrt{6})$$

$$\therefore P(x) = 2\left(x + \tfrac{1}{2}\right)(x - 1 + \sqrt{6})(x - 1 - \sqrt{6})$$

Fully Factorised

Special cases: sums and differences of cubes

In general, if $P(x) = x^3 - a^3$, then $x - a$ is a factor and so by division:

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2)$$

If a is replaced by $-a$, then

$$x^3 - (-a)^3 = (x - (-a))(x^2 + (-a)x + (-a)^2)$$

This gives:

$$x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Example 14

Factorise $x^3 - 27$.

$$x^3 - 3^3 = (x - 3)(x^2 + 3x + 9)$$

Example 14

Factorise $x^3 - 27$.

Let $P(x) = x^3 - 27$

Factor Theorem gives $P(3) = 0 \Rightarrow (x-3)$ is a factor of $P(x)$

$$
\begin{array}{r|l}
 & \quad x^2 + 3x + 9 \\
\cline{2-2}
x - 3 & x^3 + 0x^2 + 0x - 27 \\
 & -)\ x^3 - 3x^2 \\
\cline{2-2}
 & \qquad 3x^2 + 0x \\
 & \qquad -)\ 3x^2 - 9x \\
\cline{2-2}
 & \qquad\qquad 9x - 27 \\
 & \qquad\qquad -)\ 9x - 27 \\
\cline{2-2}
 & \qquad\qquad\qquad 0
\end{array}
$$

Hence, $P(x) = (x-3)(x^2 + 3x + 9)$

Section summary

- Remainder theorem When $P(x)$ is divided by $\beta x + \alpha$, the remainder is $P\left(-\frac{\alpha}{\beta}\right)$.
- Factor theorem
  - If $\beta x + \alpha$ is a factor of $P(x)$, then $P\left(-\frac{\alpha}{\beta}\right) = 0$.
  - Conversely, if $P\left(-\frac{\alpha}{\beta}\right) = 0$, then $\beta x + \alpha$ is a factor of $P(x)$.
- A cubic function can be factorised using the factor theorem to find the first linear factor and then using polynomial division or the method of equating coefficients to complete the process.
- Rational-root theorem Let $P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$ be a polynomial of degree n with all the coefficients a_i integers. Let α and β be integers such that the highest common factor of α and β is 1 (i.e. α and β are relatively prime). If $\beta x + \alpha$ is a factor of $P(x)$, then β divides a_n and α divides a_0.
- Difference of two cubes: $x^3 - a^3 = (x - a)(x^2 + ax + a^2)$
- Sum of two cubes: $x^3 + a^3 = (x + a)(x^2 - ax + a^2)$

Assigned Tasks

- **Cambridge Methods Maths:**
- Exercise 7C (All 12 Qs)